

FreeCAD Checklist: Importing a STEP File, Moving It to the Origin, and Creating an Enclosure

Version: FreeCAD 1.1.3

Part 1 – Import the STEP File

- ☐ Open FreeCAD.
 - ☐ Select the **Part Design** workbench.
 - ☐ Click **New Document** to create a new file.
 - ☐ Go to **File** → **Import** and select your **.step** or **.stp** file containing the PCB.
 - ☐ Turn on the axis cross: **View** → **Toggle Axis Cross**
 - ☐ Zoom out or adjust the view so that both the PCB and the axis cross are visible.
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Part 2 – Create a Reference Sketch for Alignment

- ☐ Make sure that no object is selected.
 - ☐ Click **New Sketch**.
 - ☐ Choose the **XY-plane** and click **OK**.
 - ☐ Draw a rectangle with the same dimensions as the PCB. For example: **60 × 50 mm**
 - ☐ Use constraints to fully define the rectangle.
 - ☐ Centre the rectangle around the origin using **Constrain Symmetry**.
 - ☐ Click **Close** to finish the sketch.
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Part 3 – Align the PCB to the Sketch

- ☐ In the **Model Tree**, select the **Sketch** containing the rectangle.
 - ☐ Hold **Ctrl** and select the imported STEP file.
 - ☐ Go to: **Edit** → **Align To** (the PCB should appear on the left side of the screen).
 - ☐ The alignment tool will display the selected objects.
 - ☐ Click three matching corner points on the rectangle and the PCB to define the alignment.
 - ☐ Right-click and choose **Align**.
 - ☐ The PCB should now be aligned with the reference sketch and positioned relative to the origin.
 - ☐ Delete the temporary alignment Sketch from the Model Tree when you are finished.
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Part 4 – Create a Body and SubShapeBinder

- ☐ Make sure that the new Body is active.
 - ☐ In the Model Tree, select the imported STEP file.
 - ☐ From the **Part Design** menu, select: **Sub-Shape Binder** or **Create SubShapeBinder**, depending on how the command is displayed in your FreeCAD interface.
 - ☐ A SubShapeBinder will be created inside the active Body.
 - ☐ Hide the original STEP file by selecting it and pressing the **Spacebar**.
- Note:** The SubShapeBinder is used as reference geometry. It allows geometry from the imported PCB to be referenced while creating features inside the Body.
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Part 5 – Create the Enclosure Base

- ☐ Select the **Body**.
- ☐ Create a **New Sketch**.
- ☐ Choose the **XY-plane** and click **OK**.
- ☐ Draw a rectangle around the PCB. For example: **64 × 54 mm**
This allows clearance around a PCB measuring approximately 60 × 50 mm.
- ☐ Centre the rectangle around the origin using **Constrain Symmetry**.
- ☐ Use **External Projection** to select the mounting-hole circles from the SubShapeBinder.
- ☐ Select each required mounting-hole circle.

Important for FreeCAD 1.1.3:

When the mounting-hole circles are selected using **External Projection**, they are already added to the Sketch as reference geometry.

Do not draw a second set of circles over the external circles.

- ☐ Check that the Sketch contains:
 - The outer rectangle.
 - The mounting-hole circles added as external/reference geometry.
- ☐ Click **Close** to finish the Sketch.
- ☐ Select the **Task** tab and then **Pad**.
- ☐ Enter the required height. For example: **15 mm**
- ☐ Click **OK**.

Note: If FreeCAD reports “**Result has multiple solids**”, check that you have not drawn duplicate circles on top of the circles brought in using External Geometry.

Apply Thickness

- ☐ Select the large top face of the Pad.
- ☐ Choose **Thickness** from the Part Design dress-up tools.
- ☐ Set the required wall thickness. For example: **1 mm**
- ☐ Make sure the top face is selected as the face to remove.
- ☐ Check the preview.
- ☐ Adjust the thickness direction if necessary so that the enclosure walls are created in the required direction.
- ☐ Click **OK**.

The result should be an open enclosure with a base and walls.

Part 6 – Create Shortened Pillars for the Mounting Holes

- ☐ Select the top face of the enclosure rim.
- ☐ Click **New Sketch**.
- ☐ Click **Toggle Construction Geometry** to use construction geometry.
- ☐ Use **External Projection** to reference the mounting-hole positions, to pick up the circle centers.
- ☐ Click **Toggle Construction Geometry** to use geometry again.
- ☐ Draw circles (slightly larger) at the mounting-hole positions to define the areas that will be used to modify the mounting pillars.
- ☐ Click **Close** to finish the Sketch.
- ☐ Select **Pocket**.

- ☐ Enter the required cut depth. For example: **10 mm**
 - ☐ Confirm the Pocket operation.
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Part 7 – Align the PCB to the Enclosure

- ☐ In the **Model Tree**, hide the SubShapeBinder (Binder).
 - ☐ Unhide the original imported STEP file.
 - ☐ Select the fixed object first: **Body**
 - ☐ Hold **Ctrl** and select the PCB STEP file.
 - ☐ Go to: **Edit** → **Align To**
 - ☐ The alignment tool will display the selected objects (PCB on the left).
 - ☐ Rotate the view so that you can clearly see the mounting holes on the PCB from the bottom.
 - ☐ Rotate the enclosure view so that you can clearly see the corresponding mounting holes inside the enclosure.
 - ☐ Select three corresponding mounting-hole circles or points on both objects.
 - ☐ Make sure that the selected locations correspond correctly.
 - ☐ Right-click and choose **Align**.
 - ☐ The PCB should now be positioned correctly inside the enclosure.
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Part 8 – Final Checks

- ☐ Verify that the PCB fits correctly inside the enclosure.
 - ☐ Check that the mounting holes align correctly with the PCB.
 - ☐ Check the enclosure wall thickness.
 - ☐ Check the base thickness.
 - ☐ Check that there is sufficient clearance between the PCB and the enclosure walls.
 - ☐ Save the FreeCAD project as: `.FCStd`
 - ☐ Optional: Export the finished enclosure as an appropriate manufacturing format.
For example:
 - **STL** for 3D printing.
 - **STEP** for CAD or manufacturing workflows.
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